

Evolutionary conservation, positive selection and structural classification: an integrative analysis of EDAR

Tran, Nhat Linh

1. Background

The EDAR gene (Ectodysplasin A Receptor) encodes a receptor of the TNF receptor family and regulates the development of ectodermal organs such as hair, teeth and sweat glands. The variant V370A is one of the best-studied signatures of positive selection in the human genome and exhibits significantly elevated allele frequencies in East Asian populations. It is associated with increased hair thickness and higher sweat gland density. These observations highlight the functional importance and evolutionary significance of EDAR.

To understand how gene structure and function evolve and how genetic variation contributes to phenotypic diversity across species, the EDAR represents a suitable model for this analysis.

2. Working Hypothesis

How can an amino acid substitution in an evolutionarily conserved death domain be positively selected without disrupting its core function?

The death domain of EDAR is conserved across vertebrates and functionally indispensable. Nevertheless, V370A demonstrates that positive selection can act within this conserved domain. It is hypothesized that the Val to Ala substitution at position 370 is structurally tolerable and functionally relevant by modulating EDAR signaling strength and is positively selected in East Asian populations due to a phenotypic advantage in ectodermal development without disrupting the core function of the death domain.

3. Planned Approach

The analysis would be conducted in three sequential steps:

Step 1: Comparative Genomic

Goal: To generate a full conservation profile of the EDAR protein and reveals which regions or positions are functionally constrained across evolutionary time.
(Where is the gene highly conserved and where are variations?)

- Dataset of Species: ~20 vertebrates covering different phenotypic types

Group	Species	Phenotyp/role
Primates	Homo sapiens + V370A, Pan troglodytes, Gorilla gorilla	close relatives

Hairy mammals	Ursus arctos, Bos mutus, Sus scrofa	dense fur -> EDAR likely active
Hairless mammals	Heterocephalus glaber, Loxodonta africana, Neophocaena asiaeorientalis, Trichechus manatus	reduced hair/hairless -> EDAR likely weaker
Birds	Struthio camelus, Dromaius novaehollandiae, Anser cygnoides, Columba livia	feathers instead of hair
Reptiles	Alligator mississippiensis, Chrysemys picta, Python bivittatus	scales
Fish	Danio rerio, Cyprinus carpio, Oncorhynchus clarkii lewisi, Scyliorhinus canicula	outgroup

- Sequences retrieved from UniProt/Ensembl/NCBI for Multiple Sequence Alignment
- Multiple Sequence Alignment using MAFFT
- Conservation score analysis and Visualization using Rate4Site/Shannon Entropy
- Synteny analysis for orthologs confirmation
- Phylogenetic relationships
- dN/dS analysis shows whether the evolutionary changes are neutral, constrained or adaptive

Position 370 will be explicitly placed in to show if it's an outlier or typical for this domain. This output is then needed for step 2.

Step 2: Population Genetics

Goal: To quantify selection signals across human populations of a curated set of EDAR variants and to correlate these signals with the output in step 1.
(Which EDAR variant show selection signals in human population?)

- Variant set: Selection of curated EDAR variants across clinically described variants, frequent population variants and known selection signatures as positive control
- Variants retrieved from 1000 Genomes Project and HGDP
- Allele frequency calculation across population (e.g. with PLINK)
- Subset of selected candidates to calculate different scores to assess selection signs
e.g. with FST (for population differentiation) and iHS (within population selection)
- Analysis of the relationship between evolutionary conservation and selection signals

The variant V370A in the EDAR gene serves as a reference (positive control), which shows strong differences in allele frequency between population and a clear signal of selection in East Asia populations.

Step 3: Structural Analysis

Goal: To explain why the conserved and selected variants identified are functionally relevant by mapping them onto the 3D structure of the protein
(Does the structural location explain why they are relevant?)

- EDAR protein structure retrieved from AlphaFold database
- Map variants from curated set and important/conserved domains (TNFR domain, Death Domain) onto 3D structure using PyMOL
- $\Delta\Delta G$ analysis to confirm structural tolerance

Position 370 is mapped onto the protein structure, which appears to lie in a functionally important region. The structure may explain the positive selection without causing diseases.

The approach to combine conservation and selection allows us a more reliable understanding of the functional importance of V370A. The structural analysis helps us to explain how it may affect the protein.